Software Requirements Specification and Design

for

LifeDoc

Personal Life Document and Deadline Management System

Version 1.0 Approved

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Revision History

Name	Date	Reason for Change	version
Anshika	24-01-2026	Initial draft of SRS document	1.0
Aditya	25-01-2026	Added Functional Requirements	1.1
Ashi	26-01-2026	Added Non Functional Requirements	1.2

1 Introduction

1.1 Purpose

To help users securely store their important documents and never miss critical deadlines by providing organized storage, smart expiry reminders, and emergency family access. To prevent loss of important personal documents To avoid missed renewals and penalties (insurance, license, passport, etc.) To keep all life-related documents in one safe digital place To support users and their families during emergencies with quick access to records

1.2 Document Conventions

• Requirements are numbered sequentially within each section • Functional requirements are prefixed with "REQ" • Terms defined in Appendix A: Glossary are italicized on first use

1.3 Intended Audience and Reading Suggestions

Intended Audience

This Software Requirements Specification (SRS) document is intended for the following audiences:

Project Supervisor and Faculty Evaluators: To understand the scope, objectives, and requirements of the LifeDoc system.

Developers: To use as a reference for system design, implementation, and testing.

Testers: To verify that the system meets the specified functional and non-functional requirements.

End Users: To gain a general understanding of the system features and usage.

System Administrators: To understand system configuration, security, and maintenance requirements.

Reading Suggestions

The document is organized to support different types of readers:

Managers and Faculty: Should read the Introduction and Overall Description sections to understand system goals and scope.

Developers: Should read the Overall Description, System Features, and Functional Requirements sections in detail.

Testers: Should focus on the Functional Requirements and Non-Functional Requirements sections.

End Users: May refer to the System Features section for a high-level overview of the application

1.4 Product Scope

LifeDoc is a personal life-document and deadline guardian system designed to solve real-life problems related to document mismanagement and missed deadlines. The system provides a centralized digital vault where users can securely store important personal documents such as identity proofs, certificates, medical records, insurance papers, and property documents.

In addition to storage, LifeDoc offers an intelligent deadline management system that tracks expiry and renewal dates of documents and notifies users in advance. The application also includes an emergency access feature that allows trusted family members to access critical documents during emergencies.

The primary goal of LifeDoc is to reduce financial loss, legal issues, and emotional stress caused by forgotten documents and deadlines. Unlike generic cloud storage platforms, LifeDoc focuses specifically on life-critical document organization and proactive reminder services.

1.4.1 Product Scope of LifeDoc

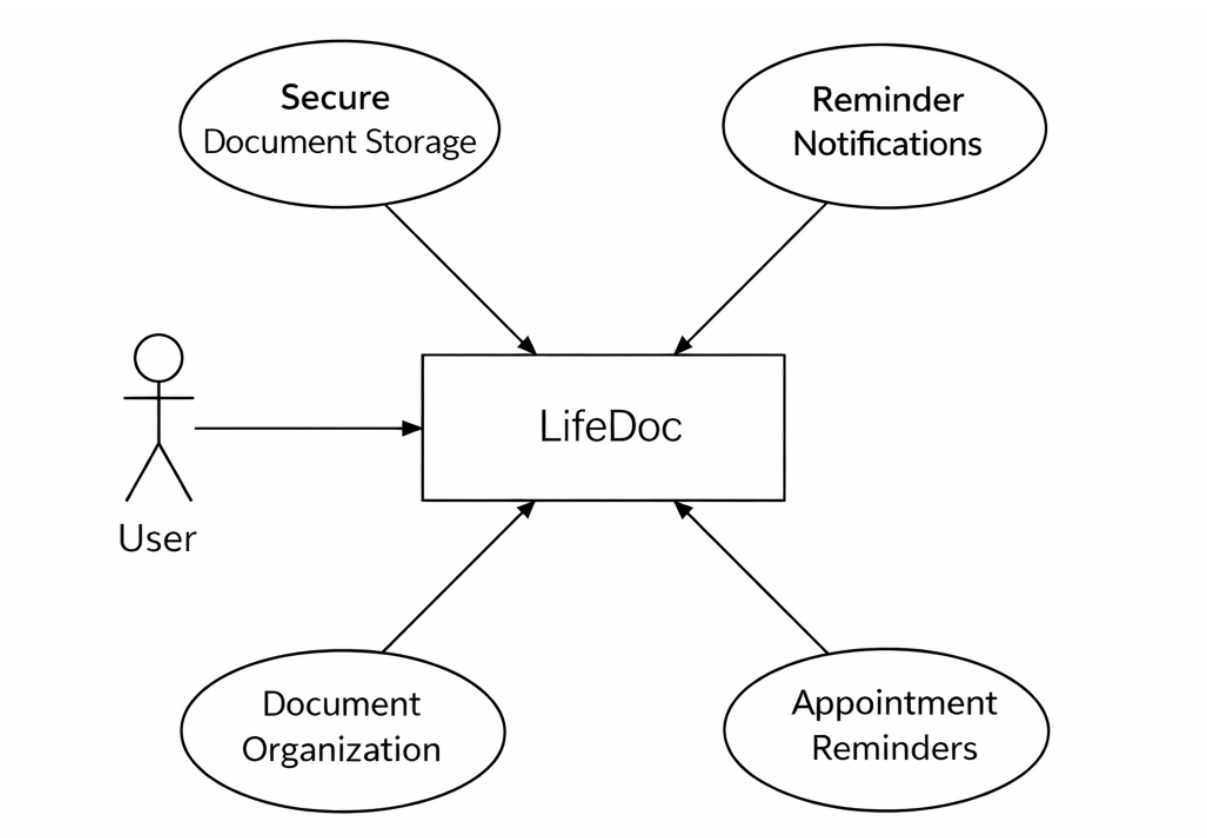

Figure 1: LifeDoc Use Case Diagram

1.5 References

• IEEE 830-1998: Recommended Practice for Software Requirements Specifications • OWASP Top 10 Security Risks • REST API Design Guidelines • React.js Documentation • Node.js Best Practices

2 Overall Description

2.1 Product Perspective

LifeDoc is a standalone web-based application designed to provide secure storage and intelligent management of personal documents and their expiry dates. The system is developed as a centralized platform where users can upload, organize, and track important documents such as identity proofs, medical records, insurance policies, and educational certificates.

LifeDoc does not replace existing government or cloud storage systems; instead, it complements them by offering value-added features such as automated expiry reminders, category-based organization, and emergency family access. The application integrates with a relational database for data storage and uses notification services for sending reminder alerts via email or SMS.

From an architectural perspective, LifeDoc follows a client–server model, where the user interacts through a web interface and all business logic and data management are handled on the server side. The system is designed to be scalable so that additional features such as OCR-based expiry detection and mobile application support can be incorporated in future versions..

2.2 Product Functions

The major functions provided by the LifeDoc system include:

- Secure upload and storage of personal documents in digital format.
- Category-wise organization of documents such as identity, education, health, finance, travel, and property.
- Setting and managing expiry or renewal deadlines for documents and services.
- Automated reminder notifications prior to important deadlines.
- Emergency access feature for trusted family members.
- Keyword-based search and instant document retrieval.

- Multi-user family dashboard to manage documents collectively.

These functions collectively ensure that users are protected from document loss, missed deadlines, and emergency unavailability of important records.

2.3 User Classes and Characteristics

The LifeDoc system is intended to be used by multiple categories of users with varying levels of technical knowledge and access privileges.

2.3.1 Primary Users (General Users)

These are individual users who store and manage their personal documents using LifeDoc.

Characteristics:

Basic knowledge of smartphones or web applications.

May belong to any age group.

Require a simple and user-friendly interface.

Expect secure storage and timely reminder notifications.

2.3.2 Family Members (Emergency Access Users)

These users are granted limited access by the primary user during emergencies.

Characteristics:

Access is restricted to selected documents only.

May not be technically advanced.

Use the system mainly for viewing and downloading documents when required.

2.3.3 System Administrator

The administrator is responsible for maintaining and managing the LifeDoc system.

Characteristics:

Has technical knowledge of system operations and databases.

Manages user accounts and system configurations.

Ensures data security, backups, and system availability.

2.3.4 Developers and Testers

These users interact with the system during development and testing phases.

Characteristics:

Have strong technical expertise.

Responsible for system maintenance, updates, and bug fixes.

Use the system to verify functional and non-functional requirements.

This classification ensures that LifeDoc addresses the needs of both technical and non-technical users while maintaining appropriate security and access control.

2.4 Operating Environment

LifeDoc is a web-based application that operates in a client–server environment. It can be accessed through any modern web browser on Windows, Linux, or macOS systems. The server runs on a standard web server with Node.js (or PHP) and a MySQL database. An active internet connection is required for system access and notification services.

2.5 Design and Implementation Constraints

The design and implementation of the LifeDoc system are subject to the following constraints:

The system shall be implemented as a web-based application using standard web technologies such as HTML, CSS, JavaScript, and a server-side programming language (Node.js or PHP).

The system shall use a relational database management system (MySQL) for storing user data and document metadata.

Due to security requirements, all user authentication and document access shall be protected using secure login mechanisms and encryption techniques.

The application must comply with basic data privacy and protection guidelines to ensure confidentiality of user documents.

The system performance may be limited by the availability of internet connectivity and server resources.

OCR-based expiry detection, if implemented, shall be dependent on the quality of uploaded document images.

The system must be compatible with commonly used web browsers without requiring specialized hardware or proprietary software.

2.6 User Documentation

LifeDoc will provide user documentation to assist users in understanding and effectively using the system. The documentation will be designed for non-technical users and will be available in digital form within the application.

The user documentation will include:

- A User Guide describing how to register, log in, upload documents, set expiry reminders, and manage family access.

- Step-by-step instructions with screenshots for key operations such as adding documents, editing expiry dates, and searching records.
- A Frequently Asked Questions (FAQ) section to address common user issues.

  Basic troubleshooting guidelines for login problems, upload errors, and notification settings.
- Security and privacy instructions to educate users on safe usage and data protection.

The purpose of the user documentation is to ensure that users can operate the LifeDoc system easily and independently without requiring technical assistance.

2.7 Assumptions and Dependencies

Assumptions:

- Users have internet access and basic knowledge of using web applications.
- Users will enter correct document and expiry information.

Dependencies:

- The system depends on server availability and database services.
- Reminder notifications depend on email/SMS services.
- Future OCR features depend on external OCR libraries and image quality.

3 External Interface Requirements

3.1 User Interfaces

LifeDoc provides a simple and user-friendly web interface that allows users to upload, organize, and manage their personal documents. The interface supports easy navigation, clear menus, and responsive design for use on both desktop and mobile devices. Important actions such as adding documents and setting reminders are accessible through intuitive buttons and forms.

3.2 Hardware Interfaces

LifeDoc does not require any specialized hardware for operation. The system can be accessed using standard devices such as desktop computers, laptops, tablets, or smartphones. Users interact with the system through a keyboard, mouse, or touch screen. The server runs on standard computing hardware capable of hosting a web server and database.

3.3 Software Interfaces

LifeDoc interfaces with standard web browsers for user interaction and communicates with a backend server for data processing and storage. It uses a relational database management system (MySQL) to store user information and document metadata. The system also integrates with email or SMS services for sending reminder notifications.

3.4 Communications Interfaces

LifeDoc uses standard internet communication protocols (HTTP/HTTPS). Users access the system through a web browser, and data is transmitted securely over the network. The system also communicates with external email or SMS services to deliver reminder notifications.

3.5 Secure Digital Vault

Description and Priority: This feature allows users to upload and securely store important personal documents. Priority: High.

Stimulus/Response Sequences:

- User selects upload option.
- System prompts for document details.
- Document is securely stored and categorized.

Functional Requirements:

- *REQ*-1: The system shall allow users to upload documents in PDF or image format.
- *REQ*-2: The system shall store documents in encrypted form.
- *REQ*-3: The system shall allow users to download stored documents.
- *REQ*-4: The system shall prevent unauthorized document access.

3.6 Smart Deadline and Reminder Management

Description and Priority: This feature tracks expiry and renewal deadlines of documents and sends reminders. Priority: High.

Stimulus/Response Sequences:

- User enters expiry date.
- System schedules reminders.

- Notification is sent before deadline.

Functional Requirements:

- *REQ*-5: The system shall allow users to set expiry dates.
- *REQ*-6: The system shall send reminder notifications before deadlines.
- *REQ*-7: The system shall support multiple reminders per document.

3.7 Emergency Family Access

Description and Priority: Allows trusted family members to access documents during emergencies. Priority: Medium.

Functional Requirements:

- *REQ*-8: The system shall allow users to assign emergency contacts.
- *REQ*-9: The system shall restrict emergency access to selected documents only.

3.8 Search and Categorization

Description and Priority: Enables fast retrieval of documents through search and categories. Priority: Medium.

Functional Requirements:

- *REQ*-10: The system shall categorize documents based on type.
- *REQ*-11: The system shall allow keyword-based document search.

4 System Features

4.1 System Feature 1: Document Management

- Upload important documents.
- Store documents securely.
- Organize documents by category.
- View and delete documents.
- Search documents by keyword.

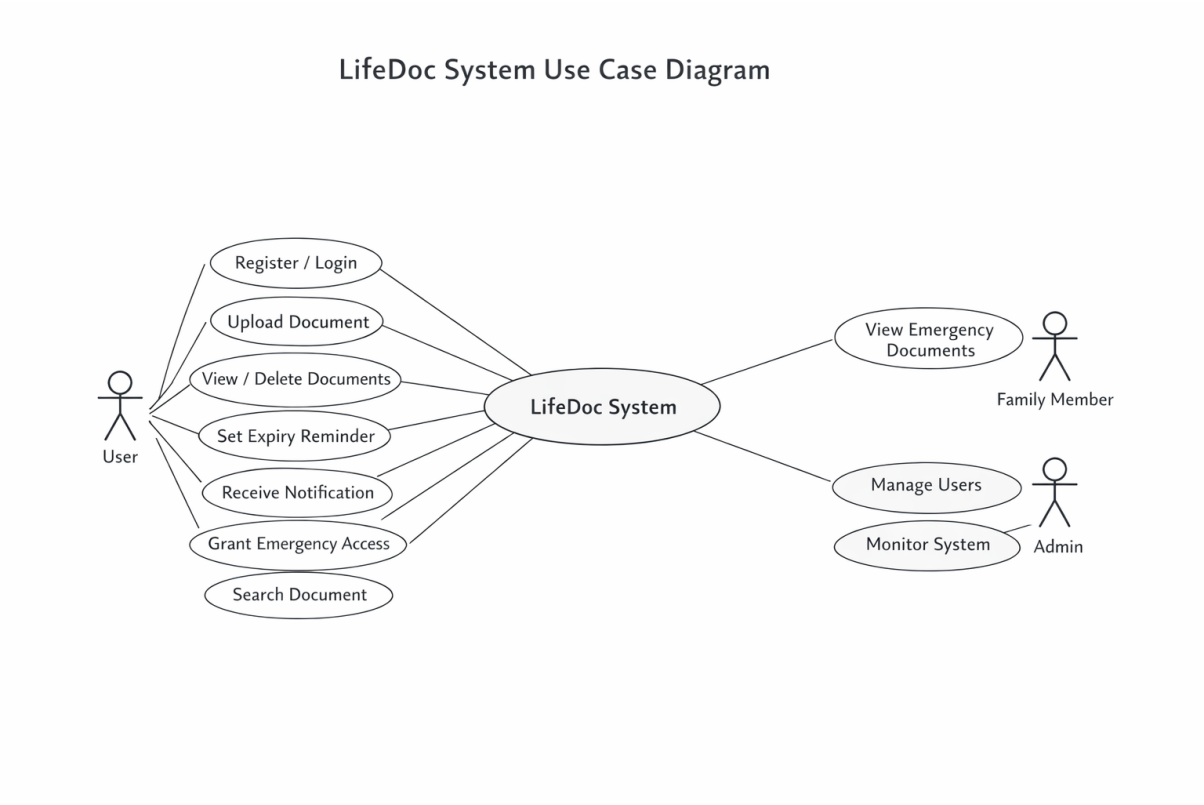

Figure 2: LifeDoc Use Case Diagram

4.2 System Feature 2: Expiry Reminder and Notification

- Enter expiry or renewal dates.
- Set reminders for documents.
- Receive alerts before expiry.
- Update reminder details.
- Cancel reminders if not needed.

4.3 System Feature 3: Emergency Family Access

- Add trusted family members.
- Grant limited document access.
- Allow view-only permission in emergencies.
- Restrict access to selected documents.
- Revoke access anytime by user.

4.4 System Feature 4: User Authentication

- Register new users.
- Login using username and password.
- Validate user credentials.
- Allow logout from the system.

  Prevent unauthorized access.

4.5 System Feature 5: Admin Management

- View all registered users.
- Monitor system activities.
- Manage user accounts.
- Handle misuse or invalid data.
- Maintain system security and stability.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall provide responses to user actions within a few seconds.
- The system shall support multiple users without significant performance loss.
- Background reminder processing shall not interrupt normal user operations.

5.2 Safety Requirements

- LifeDoc shall protect user data from unauthorized access and accidental loss.
- The system shall provide secure emergency access only to authorized family members.
- Data integrity shall be maintained even in case of system failures.

5.3 Security Requirements

- LifeDoc shall enforce a strong password policy for user accounts.
- User sessions shall automatically expire after a period of inactivity.
- User data shall be protected using encryption and secure access control.

5.4 Error Handling Requirements

At this stage of development, no explicit error handling requirements are defined for the LifeDoc system. The system shall display appropriate error messages for invalid inputs and failed operations. Detailed error handling mechanisms may be specified in future versions of the system.

5.5 Software Quality Attributes

- LifeDoc shall be usable, reliable, and secure.
- The system shall provide good performance and support future scalability.
- The software shall be easy to maintain and accessible across platforms.

5.6 Business Rules

- Users can manage only their own documents.
- Reminders are generated only for documents with expiry dates.
- Emergency access is allowed only to authorized family members.
- User data shall not be shared without consent.

6 Software Design

The Software Design section describes the overall design approach adopted for the LIFEDOC system. It provides a high-level understanding of how system components are structured and how they interact with each other. The design focuses on modularity, security, scalability, and maintainability, ensuring that the system can evolve over time without significant redesign. This section acts as a bridge between the system requirements and the implementation phase.

6.1 System Architecture

The system architecture of LIFEDOC defines the structural framework of the application and identifies the major components and their interactions. The system follows a layered architectural model, which separates responsibilities into distinct layers to improve clarity and maintainability. The primary layers include the Presentation Layer, Application Layer, and Data Layer.

The Presentation Layer handles all user interactions and provides interfaces for activities such as login, document upload, viewing documents, and receiving notifications. The Application Layer contains the core business logic of the system, including authentication processing, document management, expiry tracking, and notification scheduling. The Data Layer is responsible for storing and managing persistent data such as user details, document metadata, and system logs.

This layered architecture ensures that changes in one layer do not significantly impact other layers. It also enhances security by preventing direct access to the data layer from the user interface. The architecture supports scalability, allowing future enhancements such as cloud integration or mobile extensions without major structural changes.

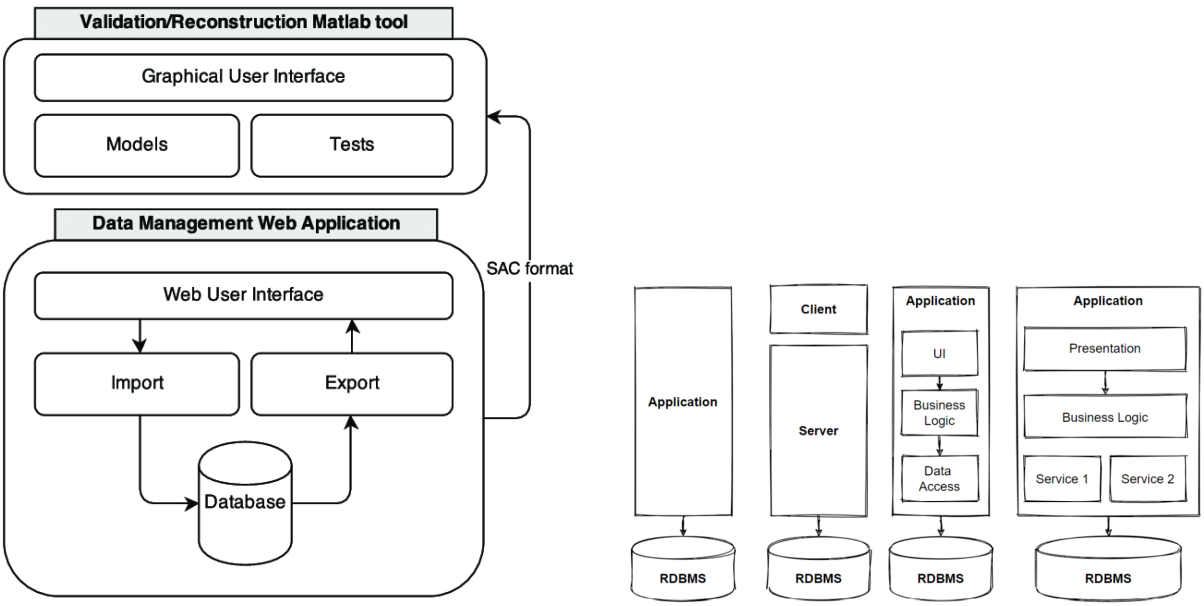

Figure 3: System Architecture Diagrams

6.2 Detailed Design

The detailed design of the LIFEDOC system provides an in-depth view of the internal structure and behavior of the system components. This design phase translates high-level architectural decisions into specific design models that guide developers during implementation. The detailed design focuses on defining system classes, their responsibilities,

interactions, and the flow of control between components.

The use of Unified Modeling Language (UML) diagrams helps visualize the system structure and behavior in a standardized manner. These diagrams improve communication among stakeholders and ensure that the system design aligns with the specified requirements.

6.2.1 Class Diagrams

Class diagrams represent the static structure of the LIFEDOC system by illustrating system classes, their attributes, methods, and relationships. These diagrams help identify key entities such as User, Document, Notification, and Admin, along with their responsibilities within the system.

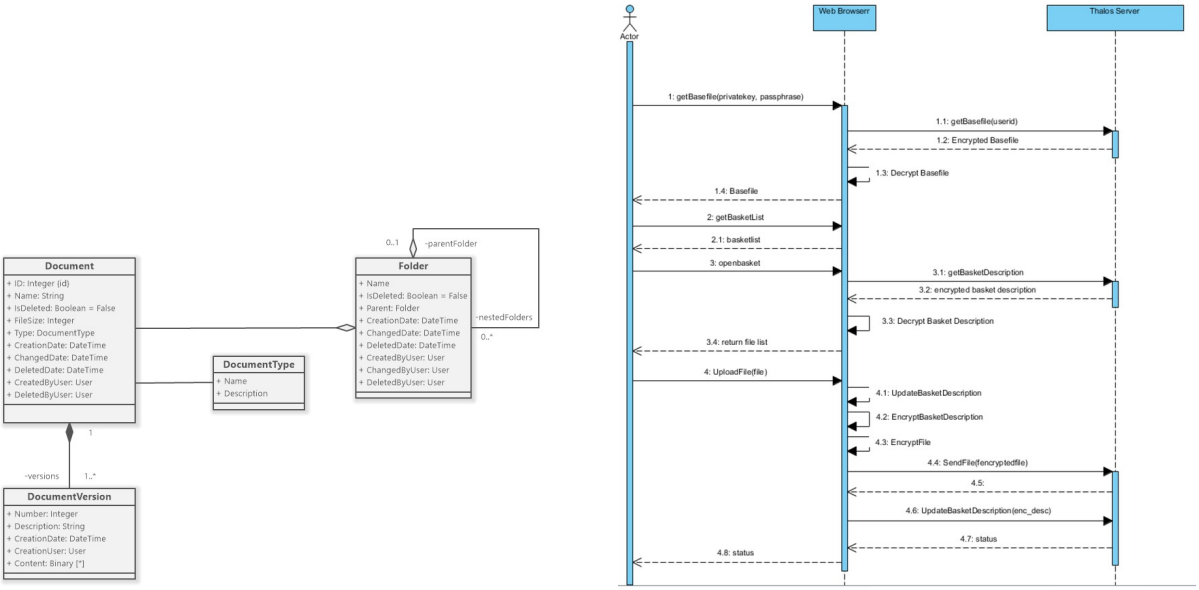

Figure 4: Class Diagrams

Each class encapsulates data and behavior related to a specific system component. Relationships such as association, inheritance, and dependency are used to represent interactions between classes. For example, a User class may be associated with multiple Document objects, while a Notification class may depend on document expiry information.

The class diagram ensures that object-oriented principles such as encapsulation, abstraction, and reusability are applied consistently. This structured representation simplifies implementation, testing, and future system enhancements.

6.2.2 Sequence Diagrams

Sequence diagrams illustrate the dynamic behavior of the LIFEDOC system by showing how objects interact over time to perform specific operations. These diagrams capture the order of messages exchanged between system components for scenarios such as user login, document upload, and expiry notification generation.

Each sequence diagram begins with a user action, followed by interactions between the Presentation Layer, Application Layer, and Data Layer. The diagrams clearly depict method calls, responses, and control flow, helping developers understand how different components collaborate to fulfill a requirement.

Sequence diagrams are particularly useful for identifying logical errors, missing interactions, or inefficient communication paths during the design phase. They also serve as a reference during debugging and testing.

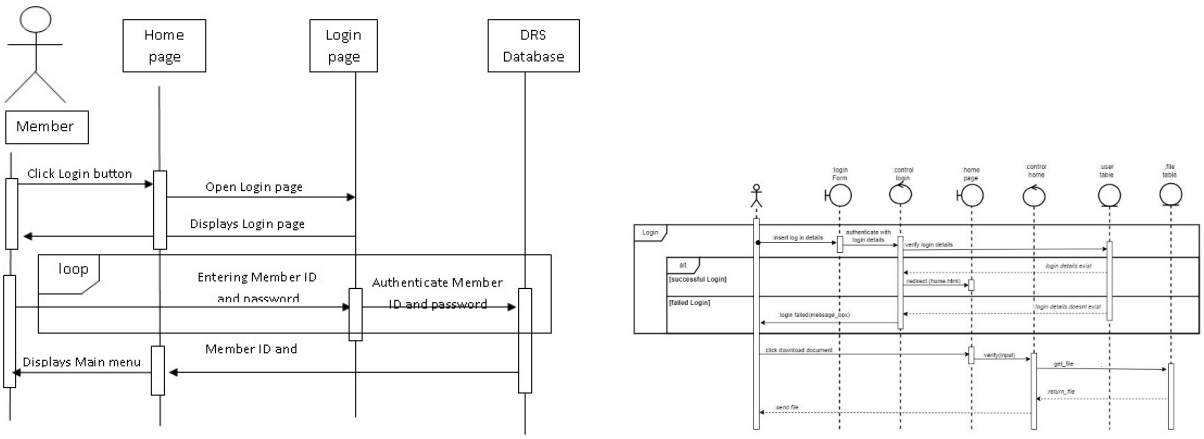

Figure 5: UML Sequence Diagram for User Login and Document Upload

6.3 Database Schema

The database schema defines the structure of data storage for the LIFEDOC system. It specifies the organization of tables, attributes, primary keys, and relationships required to store user information, documents, expiry details, and system logs. The schema is designed to ensure data integrity, consistency, and efficient retrieval.

Key entities in the database include User, Document, DocumentCategory, Notification, and AuditLog. Relationships are established using foreign keys to maintain referential integrity. Normalization techniques are applied to minimize redundancy and improve data consistency.

The database design supports scalability and secure data access. Indexing strategies are employed to optimize search and retrieval operations, particularly for document lookup and expiry tracking functionalities.

6.4 Security Design

Security design is a critical aspect of the LIFEDOC system due to the sensitive nature of personal documents. The system incorporates multiple layers of security to protect user data from unauthorized access, misuse, and data breaches. Security considerations are integrated into every layer of the system architecture.

The design emphasizes secure authentication, controlled authorization, encrypted data storage, and secure communication protocols. These measures ensure confidentiality, integrity, and availability of system data.

6.4.1 Authentication and Authorization Mechanisms

Authentication mechanisms are implemented to verify the identity of users before granting access to the system. Users are required to log in using valid credentials, and additional verification methods such as One-Time Passwords (OTP) may be used to enhance security.

Authorization mechanisms determine the actions a user is permitted to perform after successful authentication. Role-based access control is applied to differentiate between general users and administrators. This ensures that users can only access their own documents, while administrative functions remain restricted.

These mechanisms prevent unauthorized access and ensure accountability within the system.

6.4.2 Security Protocols

Security protocols define the rules and standards used to protect data during transmission and storage. The LIFEDOC system uses secure communication protocols such as HTTPS to encrypt data exchanged between clients and servers. This prevents interception and tampering during data transmission.

The LIFEDOC system follows well-defined security protocols to ensure the confidentiality, integrity, and availability of user data. Secure communication protocols such as HTTPS are used to protect data exchanged between the client and the server, preventing unauthorized interception and tampering during transmission. In addition, encryption techniques are applied to safeguard sensitive information stored within the system. All critical user data, including personal documents and credentials, is encrypted before being stored in the database, ensuring protection even in the event of unauthorized access to storage resources.

To further enhance security, secure hashing mechanisms are employed for password storage so that user credentials are never stored in plain text. Access to encrypted data is strictly controlled through authentication and authorization mechanisms, ensuring that only authorized users can view or modify their documents. These combined security

protocols provide a robust defense against common security threats and help maintain user trust in the LIFEDOC system.

Stored data, including documents and sensitive user information, is protected using encryption techniques. Regular security audits and updates are incorporated into the system design to address potential vulnerabilities. These protocols collectively ensure a secure and trustworthy system environment.

7 Traceability

Traceability ensures that every requirement specified in the Software Requirements Specification is properly addressed in the system design. It provides a clear relationship between requirements and corresponding design artifacts, ensuring that no requirement is overlooked during the design and implementation phases. Traceability also helps in impact analysis, verification, validation, and future maintenance of the system.

The traceability mechanism adopted for the LIFEDOC system maps functional and non-functional requirements to specific design components such as class diagrams, sequence diagrams, and security design elements. This mapping guarantees consistency between system requirements and design decisions.

7.1 Requirement-to-Design Mapping

This subsection presents a Requirement-to-Design Traceability Matrix (RTM) for the LIFEDOC system. The table illustrates how each requirement or use case is linked to one or more design elements. The mapping ensures that all system requirements are supported by appropriate design artifacts and that each design component can be traced back to a specific requirement.

Such traceability improves system reliability and simplifies validation during testing, as testers can directly verify whether each requirement has been implemented correctly in the design.

Requirement / Use Case ID	**Requirement / Use Case Name**	**Design ID(s)**	**Design Artifact**
UC-01	Search Document	D-CL-01, D-SQ-02	Class Diagram, Sequence Diagram
UC-05	Assign Category	D-CL-03, D-SQ-05	Class Diagram, Sequence Diagram
FR-07	Upload and Publish Document	D-CL-06	Class Diagram
NFR-02	Secure Access Control	D-SEC-01	Security Design List

A Other Requirements

A.1 Glossary

- OTP – One Time Password
- REQ - Requirements
- UC - Use Case
- FR - Functional Requirement
- NFR - Non-Funtional Requirement
- ID - Identifier
- D - Design
- CL - Class Diagram
- SQ - Sequence Diagram
- SEC - Security Design
- RTM - Requirement Traceability Matrix
- UML - Unified Modeling Language

A.2 To Be Determined List

OCR-based expiry detection.